

TELCO CUSTOMER CHURN ANALYSIS

Power BI Dashboard Documentation

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Dashboard Overview

This Power BI dashboard was built for the executive leadership team to monitor customer churn, explore customer behaviour, and support retention decisions. It is a 4-page interactive report (plus one drill-through page) covering executive KPIs, customer demographics, service-level churn drivers, and revenue impact, connected through synced slicers and page navigation buttons.

Dataset Description

Source: Telco Customer Churn Dataset (Kaggle, IBM sample dataset). The table contains one row per customer (7,043 records) with fields covering demographics (gender, senior citizen, partner, dependents), account information (tenure, contract type, payment method, monthly/total charges), subscribed services (internet service, phone service, multiple lines, online security, device protection, tech support, streaming TV/movies), and the churn outcome (Yes/No).

Data preparation

- Corrected data types.
- No duplicate customer records found; customer ID confirmed unique.
- Categorical values standardized.
- A calculated Tenure Group (binned) column was added to support the tenure-based visuals.

Data Model

Single flat fact table (one customer per row); no separate dimension tables were required given the dataset's grain. All measures reference this table directly.

Relationships: No relationship was needed (single-table model).

Dashboard Pages & Visual Descriptions

Page	Purpose	Key Visuals
Executive Overview	High-level KPI snapshot for leadership	KPI cards (Total/Churned Customers, Average Revenue, Churn Rate, Average Tenure, Retention Rate); Gender (Donut chart). Monthly Charges vs Tenure (Scatter chart). Churned by Tenure Group & Payment Method (bar chart); Churned by Contract Type (column chart); Internet Service (slicer).
Customer Insight	Demographic and account profile of churned vs. total customers	Tenure & Contract (matrix table); Senior Citizen, Dependents, Gender, Partner (bar/donut charts); Contract & churn cross-tab; Total Customers by Contract Type
Service Analysis	Identifies which services/features associate with churn	Device Protection, Streaming Movies/TV, Tech Support, Online Security (bar charts). Internet Service (treemap). Churn Drivers (decomposition-style flow) by payment method

Revenue Analysis	Quantifies financial impact of churn	Total/Lost Revenue (KPI cards); Revenue by Contract Type; Revenue Lost to Churn by Contract Type; Average Monthly Charges by Churn Status; Revenue by Payment Method; Revenue by Tenure Group & Contract (stacked column); Revenue Lost to Churn by Internet Service (Waterfall chart); Churn (slicer)
Customer Details (drill-through)	Row-level detail for any customer or segment selected on other pages	Detail table drill-through, accessible from summary visuals

Interactivity

- **Slicers:** Payment Method, Internet Service, and Churn status slicers, plus contract/tenure cross-filtering from tables and charts.
- **Navigation:** Consistent left-hand navigation buttons across all pages (Executive Overview, Customer Insight, Service Analysis, Revenue Analysis, Customer Details).
- **Drill-through:** Customer Details page, reached from summary visuals to inspect individual customer records.
- **Bookmarks:** "Reset Filter" button restores the default (unfiltered) view of the Executive Overview page.

Business Assumptions

- Churn is treated as a binary, point-in-time snapshot (customer churned or not) rather than a time-series event, consistent with the source dataset.
- Revenue figures are based on Monthly/Total Charges fields as provided; no adjustment was made for discounts, taxes, or promotions.
- "No internet service" and "No phone service" are treated as distinct categories from "No", since they represent customers who are structurally ineligible for that add-on rather than customers who opted out.

Limitations

- The dataset is a single historical snapshot with no date field, so churn cannot be trended over time or seasonally analyzed.
- Churn drivers identified here are correlational, not causal; deeper statistical or predictive modeling (e.g., logistic regression) would be needed to confirm causation.
- Customer Lifetime Value is an estimated measure derived from available fields (tenure and monthly charges), not an audited financial figure.